C++ User-Defined Function Practice

Level 1 — Basic Functions

1. Print a Message

Write a C++ program that defines a function named sayHello() that prints:

Hello, Welcome to C++ Programming!

Call the function from main().

2. Print Your Name

Write a function named printName() that takes no parameters and prints your name. Call the function from main().

3. Add Two Numbers

Write a function named add() that takes two integers as parameters and returns their sum.

Example:

Input: 10 20
Output: Sum = 30

4. Find the Square

Write a function named square() that takes an integer as a parameter and returns its square.

Example:

Input: 5
Output: Square = 25

5. Check Even or Odd

Write a function named isEven() that takes an integer and returns whether the number is even or odd.

Example:

Input: 8
Output: Even

Level 2 — Functions with Conditions

6. Find the Largest of Two Numbers

Write a function named findLargest() that takes two integers and returns the larger number.

Example:

Input: 15 27
Output: Largest = 27

7. Find the Largest of Three Numbers

Write a function named findLargest() that takes three integers and returns the largest number.

Example:

Input: 15 42 27
Output: Largest = 42

8. Check Positive, Negative, or Zero

Write a function named checkNumber() that takes an integer and prints whether the number is:

- Positive
- Negative
- Zero

9. Check Voting Eligibility

Write a function named isEligible() that takes a person's age and returns whether they are eligible to vote.

Assume the minimum voting age is **18**.

10. Find Absolute Value

Write a function named absoluteValue() that takes an integer and returns its absolute value.

Example:

Input: -15
Output: 15

Level 3 — Functions with Loops

11. Print Numbers from 1 to N

Write a function named printNumbers() that takes N as a parameter and prints all numbers from 1 to N.

Example:

Input: 5
Output: 1 2 3 4 5

12. Find the Sum from 1 to N

Write a function named calculateSum() that takes N and returns the sum of numbers from 1 to N.

Example:

Input: 5
Output: Sum = 15

13. Print Multiplication Table

Write a function named multiplicationTable() that takes a number and prints its multiplication table from 1 to 10.

Example:

Input: 5

5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50

14. Find Factorial

Write a function named factorial() that takes an integer N and returns its factorial.

Example:

Input: 5
Output: Factorial = 120

15. Count Digits

Write a function named countDigits() that takes an integer and returns the number of digits.

Example:

Input: 12345
Output: Number of digits = 5

Level 4 — Functions with Number Problems

16. Find the Sum of Digits

Write a function named sumOfDigits() that takes an integer and returns the sum of its digits.

Example:

Input: 1234
Output: Sum = 10

17. Reverse a Number

Write a function named reverseNumber() that takes an integer and returns the reversed number.

Example:

Input: 12345
Output: Reverse = 54321

18. Check Prime Number

Write a function named isPrime() that takes an integer and returns whether the number is prime.

Example:

Input: 17
Output: Prime

19. Find the Largest Digit

Write a function named largestDigit() that takes an integer and returns the largest digit.

Example:

Input: 58321
Output: Largest digit = 8

20. Check Palindrome Number

Write a function named isPalindrome() that takes an integer and checks whether it is a palindrome.

Example:

Input: 1221
Output: Palindrome

Level 5 — Functions with Arrays

21. Print Array Elements

Write a function named printArray() that takes an array and its size and prints all elements.

22. Find Array Sum

Write a function named arraySum() that takes an array and its size and returns the sum of all elements.

Example:

Input:
5
10 20 30 40 50

Output:
Sum = 150

23. Find the Largest Element

Write a function named findLargest() that takes an array and its size and returns the largest element.

24. Count Even Numbers in an Array

Write a function named countEven() that takes an array and its size and returns the number of even elements.

Example:

Input:
6
10 15 20 7 8 11

Output:
Even numbers = 3

25. Search an Element

Write a function named searchElement() that takes an array, its size, and a target value. The function should return whether the target exists in the array.

Example:

Array: 10 20 30 40 50
Target: 30

Output: Element Found